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Name:
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Date:
March 12, 2026

Abstract

This should be an overview of the student's project in less than half a page. It should summarise what they have actually done on their project: why it was done, how it was done and what were the conclusions. In particular, main objectives, main approaches/technologies used, actual solution/implementation, highlights of evaluation findings.

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Contents

1 Introduction **5**
1.1 Project Background . . . 5
1.2 Project Overview . . . 6
1.2.1 Project Outline . . . 6
1.2.2 Project Objectives . . . 7
1.3 Project Hypothesis . . . 8
1.4 Aims and Methodology . . . 9
1.5 Structure of the Report . . . 9

2 Literature and Technology Review **10**
2.1 Investigate existing mobile apps that use psychological methods to address mental health issues . . . 10
2.1.1 Prior research on mobile app interventions . . . 10
2.1.2 Discussion . . . 11
2.2 Investigate the contextualization of SA and SMA in existing apps . . . 11
2.2.1 Issues with contextualization . . . 12
2.2.2 Discussion . . . 12
2.3 Investigate psychological frameworks . . . 13
2.3.1 Existing psychological frameworks for SA and SMA . . . 13
2.3.2 Cognitive Behavioural Therapy . . . 13
2.3.3 Reality Therapy . . . 14
2.4 Investigate methods to track app usage . . . 14
2.4.1 Justification for app usage tracking . . . 14
2.4.2 Methods of app usage tracking . . . 14
2.4.3 Discussion . . . 15

3 Execution **17**
3.1 Requirements Analysis . . . 17
3.2 Design . . . 18
3.3 Implementation . . . 19
3.4 Testing . . . 20

4 Evaluation and Discussion **21**

5 Social, Legal, Ethical and Professional Issues **22**
5.1 Social Issues . . . 22
5.2 Legal Issues . . . 22
5.3 Ethical Issues . . . 22
5.4 Professional Issues . . . 22

6 Conclusions, Project Critique and Further Work **23**
6.1 Project Critique 23
6.2 Further Work 23
6.3 Conclusion 23

References **24**

1 Introduction

This section will provide a brief overview on smartphones and social media. It will then provide context on the issue of smartphone addiction and social media addiction, including how people develop such addictions, and the effects they have on an individual. Existing apps and studies that attempt to deal with the issue are also mentioned. It will also provide an overview on the project and it's objectives to be achieved in the literature & technology review and during development of the app.

1.1 Project Background

Social media (SM) and smartphones have been adopted as an important piece of daily life for a substantial number of people in the world. They have provided us the ability to connect with from around the globe on a never-before-seen scale, allowing for friendships, communities, and relationships to be built from virtually anywhere. Dating back MySpace in the mid-2000s, the release of Apple's "iPhone" and the release of the HTC Dream running the "Android" operating system in 2008, to SM such as Twitter and Facebook reaching out to billions around the globe, smartphones and social media looks to be omnipresent within a majority of peoples' lives, with it being estimated that 1 in 3 people globally have some form of SM presence (Shah 2021), reportedly 85% of adults in the United States own a smartphone (Ouellette and Rao 2022).

Over the years, worries have been raised regarding many issues that concern smartphones and SM, particularly surrounding the problem of individuals developing an overdependence on their smartphones and SM that negatively affects their day-to-day life (Allen and Buzzi 2020; Primack et al. 2017). Commonly categorised as "Smartphone Addiction" (SA) and "Social Media Addiction" (SMA) and considered subsets of behavioural addiction (King and Dong 2017; Hou et al. 2019), the issue has been largely discussed and researched over the last 15 years, with researchers looking into potential causes and correlations with other health effects.

Bhalerao et al. (2020) analyzed SA and provided potential causes behind SA, one of which being how the current generation has been brought up alongside the internet and technology, therefore possibly feeling a preference to communicate over the internet instead of in-person, fueling the dwindling need for face-to-face communication. King and Dong (2017) made a series of hypotheses in relation to college students and the effects related to SA, looking at SA's associations with "life stressors" (a measure of stress and anxiety), "digital overload" (a measure of information and communication overload), retention, academic performance, and social well-being.

The results found that SA could be positively correlated with life stressors, digital overload, retention, and academic performance, something consistent with previous studies.

Potential treatments have been researched as well. A China-based study from 2019 analyzing SMA (Hou et al. 2019) developed a cognitive-behavioural-therapy-based intervention program that used various techniques including reminder cards, journal entries before sleeping, and cognitive reconstruction. The results showed that compared to the control group, those who took part in the program saw improvements in mental health, engagement in learning, sleep quality, and also saw a drop in SM usage.

The "AntiSocial" app, developed by Australian company BugBean for Android devices allowed users to restrict access to certain apps and provided app usage statistics, but also added other functionality like usage comparisons with others that were within their age group, location, and occupation. It also showed users how many times they picked up their device, and how long they used certain apps for (Michael 2017). The author stated that the app ranked 3rd in downloads on the UK Google Play store mere days after launch, "demonstrat[ing] the need for [AntiSocial.io]" (ibid.).

However, there seemingly exists a gap of intervention apps that deal specifically with SA or SMA, and that also allow self-reflection from the user, and most literature on the topic mostly provide analysis on apps that deal with adjacent issues. A 2021 study by Oliveira et al. (2021) reviewed the effectiveness of psychological intervention mobile apps, and while they concluded that the apps showed promise, none of the literature specifically looked at addressing the issue of SA or SMA.

1.2 Project Overview

1.2.1 Project Outline

Evidence has shown that smartphone apps that help with SA and SMA are in-demand, and significant amounts research have been conducted suggesting treatments to the issue. However, most consumer products on the app store opt to utilize app-restriction features. Because of this approach, young adults who use these apps aren't given much opportunity to reflect on the nature of SA & SMA and the potential causes of it, leading to very little character growth for the end-user. However, by taking a different approach by integrating psychology, young adults can be given the opportunity to reflect on their addiction and learn how to address the

issue. With that being said, the research statement is as follows:

To create a mobile application that integrates psychological intervention methods alongside usage tracking technologies to help address smartphone and social media addiction

1.2.2 Project Objectives

The primary aim of this project is to create an application that can guide the user into reducing their smartphone & social media usage with the use of usage tracking and psychological methods.

Secondary Objectives

Investigate existing mobile apps that use psychological methods to address mental health issues
Research into how existing mobile applications make use of psychological methods when addressing issues such as depression and anxiety to understand how viable using mobile apps are to address similar issues.

Investigate the contextualization of SA and SMA in existing apps
Research how the issue of SA and SMA are contextualized within existing intervention apps and what issues could arise from how they are presented. This will help in enabling growth in a user when it comes to learning about SA and SMA.

Investigate psychological frameworks
Research how psychological methods have been used in the past, specifically to treat SA or SMA, and see if such methods have been integrated with mobile apps to gain a further understanding on how these methods could be integrated or improved upon.

Investigate methods to track app usage
Research into how apps currently track phone activity, and how they can be utilized in the context of the project, as a major component of the application makes use of usage tracking.

Primary Objectives

Develop the smartphone intervention app
Create the application based on the knowledge gained from the literature and technology review. This will involve integrating the chosen psychological framework and usage tracking method before testing the app's functionality via test cases to ensure

the app functions as intended.

Gather participants for evaluating the app and perform evaluation
Participants will be used when evaluating the developed application. They will be given an overview of the app before being given a consent form to sign to consent to participating in the research.

Analyze the results and obtain a conclusion
Collect results from the evaluation and see if the app was impactful at reducing smartphone and social media usage. Then evaluate based on the results, and see if the problem statement and hypothesis were met.

1.3 Project Hypothesis

Testing the hypothesis will involve providing participants with the developed application for a period of three weeks. During this period, participants will attempt to reduce their smartphone & social media usage by using the developed mobile application with usage tracking and psychological intervention methods. Given this, the hypothesis for this project is:

The combination of usage tracking and intervention methods rooted in psychology within the created mobile application will help to address and reduce smartphone & social media usage in young adults.

1.4 Aims and Methodology

1.5 Structure of the Report

2 Literature and Technology Review

This literature and technology review will review mobile apps that use psychological methods to address mental health issues before looking at how existing apps provide context regarding SA & SMA and the issues associated with it. Potential psychological frameworks for treating SA & SMA are reviewed before researching how app usage can be tracked.

2.1 Investigate existing mobile apps that use psychological methods to address mental health issues

The project background mentioned that although researchers have studied SA and SMA, few apps are available, or have been created for research purposes that handle the issue. However, apps that address similar issues exist and have extensively been reviewed and evaluated. This section reviews existing studies on these apps to gain a further understanding on the viability of treating mental health issues.

2.1.1 Prior research on mobile app interventions

Researchers Weisel et al. (2019) reviewed mobile apps that integrated psychological frameworks to treat mental health issues such as depression, anxiety, drug use, among others, with their intention being to evaluate the effectiveness of these apps in reducing mental health symptoms. The review discusses features used, with the most common being behaviour monitoring, participant engagement via explicit input, tailoring based on feedback, and gamification, while also noting the majority of apps opted to use CBT as their psychological framework. They discover that out of all the mental health issues targeted, depression, sleep-related issues, and smoking returned a significant effect, with the review stating that "our knowledge on how to design effective mental health apps is very much at the beginning".

They continue by suggesting different approaches that focus more on the user and the framework, as well as engagement with the app via usage patterns, before saying that at the current point in time, there isn't much significant evidence that supports using mobile apps for treating mental health as an alternative to traditional treatment, highlighting that while apps targeting depression, sleep issues and smoking showed promising results, the results showed a high risk of bias within them (ibid.).

Oliveira et al. (2021) reviewed mobile app-based interventions utilizing psychological frameworks targeting college students, aiming to review the outcomes of using

app-based psychological interventions instead of traditional therapy, as well as looking how these apps are being embraced by students. The review details the types of mental health issues these apps target, while also noting that a majority of these apps are "self-guided" in nature (Oliveira et al. 2021). Psychological frameworks used are mentioned, with Cognitive Behavioural Therapy (CBT) being the main framework used or combined with another framework, with other frameworks such as Acceptance and Commitment Therapy and Self-determination Theory mentioned.

The review then concludes by saying that the current literature demonstrates that using mobile apps for mental health intervention is a viable option and could be a great resource for counselling services, noting that future work into the topic should utilize alternative psychological frameworks, look further into what students and therapists would want out of similar apps, and that future studies should stray from monetary incentives for students looking to participate (ibid.).

2.1.2 Discussion

Weisel et al. (2019) found that while apps that targeted depression, sleep-related issues, and smoking showed some results, the field was merely in its infancy when it came to designing effective applications that integrate psychological frameworks. Later, Oliveira et al. (2021) showed that mobile apps that targets mental health issues aimed towards college-level students were a viable alternative option for treating mental health issues due to student embracement, showcasing growth within the topic between the two points in time, though noted that this could've been due to incentives attached within the latter study.

Despite lacking insight on psychological frameworks other than CBT, showing a need to expand the types of psychological frameworks used for these apps, and the use of incentives which may bias user retention, the studies demonstrate that the viability of a mobile mental health app using psychological frameworks has grown, and that developing such app is doable, as long as aspects such as other psychological frameworks, and user incentive are considered.

2.2 Investigate the contextualization of SA and SMA in existing apps

The project background mentioned that intervention apps are developed with mostly functionality in mind, while neglecting any sort of teaching element, therefore leaving the user little to no room to learn about SA and SMA. This section reviews how SA and SMA are contextualized, and how this could be improved.

2.2.1 Issues with contextualization

Dosselaar (2017) puts forward an argument stating that SA apps created by developers aren't suitable for treating SA due to developers having limited knowledge on the topic of SA, and points out the irony in attempting to treat SA with the use of a smartphone. This sentiment is shared in Petrucco (2021), stating that proposing technology as a solution to an issue caused by technology itself is paradoxical in nature.

Dosselaar says that developers lack a deep understanding of the topic, and points out the overall complexity of the issue, mentioning how SA is a product of many other factors; not just the phone, and the apps simplify what is often a complex issue by acting as a one-size-fits-all solution (Dosselaar 2017; Roffarello and Russis 2019). They also fail to consider context behind smartphone usage, noting how it can be productive (Dosselaar 2017) and non-disruptive depending on where it's used (Petrucco 2021). Another issue is that the apps lack any kind of opportunity for the user to understand and reflect on the issue of SA, noting that these apps frame SA as an issue that can be quantified, while lacking any features that offer information about SA (Dosselaar 2017), something that plays a role in treating the issue (Rosa, Stracuzzi, and Faro 2022), and could influence app effectiveness (Rahmillah et al. 2023).

The author concludes by stating that SA apps are limited in effectiveness as they focus on practical functionality of the app rather than the contextualization and opportunity for reflection for the user, before recommending researching integrating psychological frameworks into the apps (Dosselaar 2017).

2.2.2 Discussion

While it can be argued that some of these issues have been addressed, for consumers, most of these issues are yet to be addressed by anyone but the developers of these apps (ibid.). These apps focus only on features such as self-monitoring and app restriction (Löchtefeld, Böhmer, and Ganev 2013), rather than providing opportunity for end-users to reflect and learn. These apps frame SA and SMA as an issue that can be solved with the apps themselves, rather than it being a complex issue (Santhosh 2025), emphasised by the fact that not all smartphone usage is the same for each person (Petrucco 2021), bringing suggestions of incorporating aspects such as psychological frameworks into these apps to make them more effective (Dosselaar 2017).

2.3 Investigate psychological frameworks

The project background mentioned that plenty of studies have been conducted utilising different psychological methods to treat SA and SMA. This section aims to review the most common psychological methods to understand which method would work best with the project.

2.3.1 Existing psychological frameworks for SA and SMA

The previous section highlighted the need for research into how psychological frameworks could be integrated into consumer SA and SMA intervention apps to make them more effective, a sentiment shared with Oliveira et al. (2021), who mentioned future intervention apps should incorporate [other] psychological methods.

Intervention apps developed for research purposes commonly use Cognitive Behavioural Therapy (CBT) as their psychological framework (Weisel et al. 2019; Oliveira et al. 2021), though other frameworks such as mindfulness, reality therapy, and theory of planned behaviour have also been utilized to mixed success (Gajecki et al. 2014; Witkiewitz et al. 2014; Lu et al. 2025).

2.3.2 Cognitive Behavioural Therapy

Cognitive Behavioural Therapy (CBT) is a psychological framework that take an "active, problem-focused, time-sensitive treatment approach" aiming to increase adaptive behaviour and reduce emotional distress in an individual with mental health issues via targeted, short-term strategies that intends to alter maladaptive emotional responses in an individual by changing their thoughts and/or behaviours (Wenzel 2017; Kaczkurkin and Foa 2015). Despite results lacking in some areas, it can be argued that it's the framework with the highest usage (Hofmann et al. 2012) overall, with some considering it the "gold standard" (Beck 2020) due to the proven success the framework has had on treating certain issues such as nicotine overdependence, schizophrenia, depression, and anxiety (Hofmann et al. 2012).

The main CBT techniques used in intervention apps are behavioural activation (A focus on engaging a patient in positive activities and helping with aversive control (Tindall et al. 2017)), cognitive restructuring (A process that assists patients in identifying and evaluating negative thoughts (Wenzel 2017)), and problem solving (Discover solutions to a patient's issue (Denecke, Schmid, and Nüssli 2022)). Throuvala et al. (2020) assessed if these techniques could reduce smartphone usage via three separate apps which provided these techniques (AntiSocial, Headspace, Pacifica) and found that those in the experimental group saw their smartphone us-

age drop and were made more aware of their usage when compared to the control group.

2.3.3 Reality Therapy

Reality Therapy (RT) is a psychological framework that can be traced back to choice theory, which theorizes that all human behaviour is a result of choices, (Bradley 2014; Wubbolding 2016). RT's primary aim is on focusing an individual's behaviour in the present by using the Wants, Direction, Evaluation, and Planning & Commitment System (WDEP system) (Bradley 2014). This involves asking an individual to define a goal for themselves, what they're currently doing, and if that behaviour is halting progress towards their goal or not, before being encouraged to replace it with a healthier alternative (Yao et al. 2017). RT has shown effectiveness in treating issues such as Internet Addiction and Internet Gaming Disorder (Yao et al. 2017; Zajac et al. 2017), though has had limited research (Bradley 2014) when treating SA and SMA compared to CBT.

Rad and Ahmadi (2018) developed a mobile app integrating RT. The app was hypothesized to reduce anxiety, SM usage, among others, and was developed for Android devices. It implemented teachings about SMA, as well as app restriction functions, with the results being that anxiety and SM usage fell when compared to the control group.

2.4 Investigate methods to track app usage

The project background mentioned that existing SA and SMA intervention applications mainly make use of app-restriction and usage tracking technologies, and due to usage tracking being a large component of the project, it would be useful to know the technological methods used for app usage tracking.

2.4.1 Justification for app usage tracking

App usage tracking has proven to be successful in reducing smartphone usage. a study by researchers Rahmillah et al. (2023) reviewed the effectiveness of apps intended to reduce smartphone use, and found that in the 13 apps reviewed, 4 apps successfully reduced smartphone usage, those of which included app usage tracking.

2.4.2 Methods of app usage tracking

Rosa, Stracuzzi, and Faro (2022) hypothesized that individuals with high levels of smartphone usage show more difficulty in not using their smartphone when compared

to those with lower levels, verifying their hypothesis by developing an application called "SocialStatsApp" to track app usage of participants.

The app was developed for devices using Google's Android operating system. The researchers mention how making the app compatible for Apple's iOS devices wasn't possible as the OS disallows the collection of usage statistics for privacy reasons (Apple-Support n.d.). The programming language used was Java, one of three officially supported languages via the Android SDK, the other two being Kotlin and C++ (Android-Developers n.d.), though the study doesn't detail the reasoning behind using Java. The app makes use of the Android class android.app.usage.UsageStatsManager from the Android API which gives access to phone usage stats and history. Because of this, the app needs permission manually granted by the user. The results of the study showed that those with high levels of smartphone usage did experience more difficulty in not using their phone.

Bye et al. (2024) analyzed the effects technology has on the mental health of young people, developing an app called the "3S-YP" app that constantly collected "smartphone metadata" from participants. While they don't detail the development of the app, they mention that smartphone metadata was collected using the android.app.usage package from the Android API, while also mentioning that metadata collection was not performed for the iOS version of the application due to "data protection restrictions", a sentiment shared by the previous study, showing that developing for Android allowed for additional functionality not possible with iOS.

Another method of tracking a user's app usage is by making use of screenshots and digital well-being features included with smartphones. Ubochi (2019) investigates the use of digital health apps and their effectiveness via Apple's Screen Time feature. Due to the aforementioned privacy restrictions, the researcher instructs participants to instead screenshot multiple aspects related to the Screen Time feature. A limitation of this method however is that screenshots must be taken at a specified time, as mistiming it could yield less accurate metrics (*Mobile sensing in psychology: Methods and applications* 2024).

2.4.3 Discussion

Both studies chose Android as their development platform of choice, mentioning that iOS apps don't allow for app usage tracking due to privacy restrictions (Rosa, Stracuzzi, and Faro 2022; Bye et al. 2024; Apple-Support n.d.). This is elaborated further within *Mobile sensing in psychology: Methods and applications* (2024), mentioning including iOS users would require a "jailbroken" phone, which brings a variety of issues. Meanwhile, Android allows for this as long as the user permits it. Only one of the studies makes mention of a programming language (Rosa,

Stracuzzi, and Faro 2022), that being Java, and while both studies make use of the android.app.usage package within the Android API, only one (Rosa, Stracuzzi, and Faro 2022) explicitly uses the UsageStatsManager class. Given this, it is clear that the OS that the app should be developed for is Android, and that using the android.app.usage package would be a viable method of app usage tracking.

3 Execution

This section should provide a detailed description of the research methods which will used in the project. This will include the process of creating the artefact and evaluating the outcome. This should make reference to the literature review findings where appropriate.

3.1 Requirements Analysis

3.2 Design

3.3 Implementation

3.4 Testing

4 Evaluation and Discussion

This section provides discussion of the evaluation part of the project in relation to the hypothesis stated in section 1.3, which was:

The combination of usage tracking and intervention methods rooted in psychology within the created mobile application will help to address and reduce smartphone & social media usage in young adults.

5 Social, Legal, Ethical and Professional Issues

5.1 Social Issues

5.2 Legal Issues

5.3 Ethical Issues

5.4 Professional Issues

6 Conclusions, Project Critique and Further Work

6.1 Project Critique

6.2 Further Work

6.3 Conclusion

References

Allen, Ruth and Peter Buzzi (2020). "Social media and mental health social work". In: *Social Media and Social Work*. Policy Press, pp. 113–132.

Android-Developers (n.d.). *Application fundamentals — App architecture — Android Developers*. English. URL: https://developer.android.com/guide/components/fundamentals.

Apple-Support (n.d.). *If an app asks to track your activity - Apple Support (UK)*. English. URL: https://support.apple.com/en-gb/102420.

Beck, Judith S. (2020). *Cognitive behavior therapy: Basics and beyond*. Guilford Publications.

Bhalerao, Muktpal M. et al. (2020). "An analysis of smartphone addiction among MBBS students". In: *Indian Journal of Clinical Anatomy and Physiology* 7.1, pp. 1–7.

Bradley, Ezrina L. (2014). "CHOICE THEORY AND REALITY THERAPY: AN OVERVIEW". English. In: *International Journal of Choice Theory and Reality Therapy* 34.1. Source type: Scholarly Journals; Object type: Article; Object type: Feature; Copyright: Copyright International Journal of Reality Therapy Fall 2014; DOCID: 3496477601; PCID: 91773702; PMID: 165936; ProvJournalCode: NJRT; PublisherXID: INNNNJRT0009496428, pp. 6–13. URL: https://www.proquest.com/scholarly-journals/choice-theory-reality-therapy-overview/docview/1625136609/se-2?accountid=15977%20https://discover.gcu.ac.uk/discovery/openurl?institution=44GLCU_INST%5C&vid=44GLCU_INST:44GLCU_VU2%5C&?url_ver=Z39.88-2004%5C&rft_val_fmt=info:ofi/fmt:kev:mtx:journal%5C&genre=article%5C&sid=ProQ:ProQ%5C%253Apsychology%5C&atitle=CHOICE+THEORY+AND+REALITY+THERAPY%5C%253A+AN+OVERVIEW%5C&title=International+Journal+of+Choice+Theory+and+Reality+Therapy%5C&issn=%5C&date=2014-10-01%5C&volume=34%5C&issue=1%5C&spage=6%5C&au=Bradley%5C%252C+Ezrina+L%5C&isbn=%5C&jtitle=International+Journal+of+Choice+Theory+and+Reality+Therapy%5C&btitle=%5C&rft_id=info:eric/%5C&rft_id=info:doi/.

Bye, Amanda et al. (2024). "Cohort profile: The Social media, smartphone use and Self-harm in Young People (3S-YP) study–A prospective, observational cohort study of young people in contact with mental health services". English. In: *PLoS One* 19.5, e0299059. DOI: 10.1371/journal.pone.0299059.

Denecke, Kerstin, Nicole Schmid, and Stephan Nüssli (2022). "Implementation of cognitive behavioral therapy in e–mental health apps: literature review". In: *Journal of medical Internet research* 24.3, e27791.

Dosselaar, C. van (2017). "Smartphone addiction? There's an app for that!-How "smartphone addiction apps" frame smartphone addiction through discourse and affordances". PhD thesis.

Gajecki, Mikael et al. (2014). "Mobile phone brief intervention applications for risky alcohol use among university students: a randomized controlled study". In: *Addiction science & clinical practice* 9.1, p. 11.

Hofmann, Stefan G. et al. (2012). "The efficacy of cognitive behavioral therapy: A review of meta-analyses". In: *Cognitive therapy and research* 36.5, pp. 427–440.

Hou, Yubo et al. (2019). "Social media addiction: Its impact, mediation, and intervention". In: *Cyberpsychology: Journal of psychosocial research on cyberspace* 13.1.

Kaczkurkin, Antonia N. and Edna B. Foa (2015). "Cognitive-behavioral therapy for anxiety disorders: an update on the empirical evidence". In: *Dialogues in clinical neuroscience* 17.3, pp. 337–346.

King, Ruth C. and Su Dong (2017). "The impact of smartphone on young adults". In: *The Business & Management Review* 8.4, p. 342.

Löchtefeld, Markus, Matthias Böhmer, and Lyubomir Ganev (2013). "AppDetox: helping users with mobile app addiction". In: *Proceedings of the 12th international conference on mobile and ubiquitous multimedia*, pp. 1–2.

Lu, Peng et al. (2025). "Interventions for Digital Addiction: Umbrella Review of Meta-Analyses". English. In: *Journal of Medical Internet Research* 27, e59656. DOI: 10.2196/59656. URL: https://www.proquest.com/scholarly-journals/interventions-digital-addiction-umbrella-review/docview/3222368316/se-2?accountid=15977%20https://discover.gcu.ac.uk/discovery/openurl?institution=44GLCU_INST%5C&vid=44GLCU_INST:44GLCU_VU2%5C&?url_ver=Z39.88-2004%5C&rft_val_fmt=info:ofi/fmt:kev:mtx:journal%5C&genre=article%5C&sid=ProQ:ProQ%5C%253Ahealthcompleteshell%5C&atitle=Interventions+for+Digital+Addiction%5C%253A+Umbrella+Review+of+Meta-Analyses%5C&title=Journal+of+Medical+Internet+Research%5C&issn=%5C&date=2025-01-01%5C&volume=27%5C&issue=%5C&spage=e59656%5C&au=Lu%5C%252C+Peng%5C%253BQiu%5C%252C+Jiamin%5C%253BHuang%5C%252C+Shiqi%5C%253BWang%5C%252C+Xinman%5C%253BHan%5C%252C+Shasha%5C%253BZhu%5C%252C+Sui%5C%253BYoujing+Ning%5C%253BFang-fang%5C%252C+Zeng%5C%253BYuan%5C%252C+Yuan%5C&isbn=%5C&jtitle=Journal+of+Medical+Internet+Research%5C&btitle=%5C&rft_id=info:eric/%5C&rft_id=info:doi/10.2196%5C%252F59656%20https://doi.org/10.2196/59656.

Michael, Katina (2017). "Are you addicted to your smartphone, social media, and more?: The new AntiSocial app could help". In: *IEEE Consumer Electronics Magazine* 6.4, pp. 116–121.

Mobile sensing in psychology: Methods and applications (2024). English. Source type: Books; Object type: Book; Object type: Reference Document; AccNum: 2024-33139-000; CSAUnique: PSIN-2024-33139-000. New York, NY: The Guil-

ford Press, p. 802. URL: https://www.proquest.com/books/mobile-sensing-psychology-methods-applications/docview/2919441599/se-2?accountid=15977%20https://discover.gcu.ac.uk/discovery/openurl?institution=44GLCU_INST%5C&vid=44GLCU_INST:44GLCU_VU2%5C&?url_ver=Z39.88-2004%5C&rft_val_fmt=info:ofi/fmt:kev:mtx:book%5C&genre=book%5C&sid=ProQ:ProQ%5C%253Apsycinfo%5C&atitle=%5C&title=Mobile+sensing+in+psychology%5C%253A+Methods+and+applications%5C&issn=%5C&date=2024-01-01%5C&volume=%5C&issue=%5C&spage=%5C&au=%5C&isbn=9781462553105%5C&jtitle=%5C&btitle=Mobile+sensing+in+psychology%5C%253A+Methods+and+applications%5C&rft_id=info:eric/2024-33139-000%5C&rft_id=info:doi/.

Oliveira, Carla et al. (2021). "Effectiveness of Mobile App-Based Psychological Interventions for College Students: A Systematic Review of the Literature". In: *Frontiers in Psychology* olume 12 - 2021. URL: https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2021.647606.

Ouellette, Samantha and Babar K. Rao (2022). "Usefulness of smartphones in dermatology: a US-based review". In: *International Journal of Environmental Research and Public Health* 19.6, p. 3553.

Petrucco, Corrado (2021). "User and Smartphones Social and Cognitive Interaction". In: *Teaching and Mobile Learning*. CRC Press, pp. 75–92.

Primack, Brian A. et al. (2017). "Social Media Use and Perceived Social Isolation Among Young Adults in the U.S." In: *American Journal of Preventive Medicine* 53.1. ID: 271902, pp. 1–8. DOI: 10.1016/j.amepre.2017.01.010. URL: https://www.sciencedirect.com/science/article/pii/S0749379717300168.

Rad, Mahsa Esmaeili and Farid Ahmadi (2018). "A new method to measure and decrease the online social networking addiction". In: *Asia-Pacific Psychiatry* 10.4, e12330.

Rahmillah, Fety Ilma et al. (2023). "Evaluating the effectiveness of apps designed to reduce mobile phone use and prevent maladaptive mobile phone use: multimethod study". In: *Journal of medical Internet research* 25. pmid:37643002, e42541.

Roffarello, Alberto Monge and Luigi De Russis (2019). "The race towards digital wellbeing: Issues and opportunities". In: *Proceedings of the 2019 CHI conference on human factors in computing systems*, pp. 1–14.

Rosa, Angela Fabio, Alessia Stracuzzi, and Riccardo Lo Faro (2022). "Problematic Smartphone Use Leads to Behavioral and Cognitive Self-Control Deficits". English. In: *International Journal of Environmental Research and Public Health* 19.12, p. 7445. DOI: 10.3390/ijerph19127445.

Santhosh, Joseph (2025). "Augmented Digital Fotonovela (SMAART App) to Mitigate Health Consequences Among Adolescents Resulting from Social Media Ad-

diction". English. Source type: Dissertations & Theses; Object type: Dissertation; Object type: Thesis; Copyright: Database copyright ProQuest LLC; ProQuest does not claim copyright in the individual underlying works.; DissertationNum: 31839151; DocISBN: 9798288851650; PublisherXID: 31839151; PMID: 66569; MSTARLegacyID: DISS-31839151. PhD thesis. ISBN: 9798288851650. URL: https://www.proquest.com/dissertations-theses/augmented-digital-fotonovela-smaart-app-mitigate/docview/3231139889/se-2?accountid=15977%20https://discover.gcu.ac.uk/discovery/openurl?institution=44GLCU_INST%5C&vid=44GLCU_INST:44GLCU_VU2%5C&?url_ver=Z39.88-2004%5C&rft_val_fmt=info:ofi/fmt:kev:mtx:dissertation%5C&genre=dissertations%5C&sid=ProQ:ProQuest+Dissertations+%5C%2526+Theses+Global%5C&atitle=%5C&title=Augmented+Digital+Fotonovela+%5C%2528SMAART+App%5C%2529+to+Mitigate+Health+Consequences+Among+Adolescents+Resulting+from+Social+Media+Addiction%5C&issn=%5C&date=2025-01-01%5C&volume=%5C&issue=%5C&spage=%5C&au=Santhosh%5C%252C+Joseph%5C&isbn=9798288851650%5C&jtitle=%5C&btitle=%5C&rft_id=info:eric/%5C&rft_id=info:doi/.

Shah, Aastha (2021). "Rise of Social Media and Its Trends". In: *International Journal of Research Culture Society* 22, p. 162.

Throuvala, Melina A. et al. (2020). "Mind over matter: testing the efficacy of an online randomized controlled trial to reduce distraction from smartphone use". In: *International journal of environmental research and public health* 17.13, p. 4842.

Tindall, Lucy et al. (2017). "Is behavioural activation effective in the treatment of depression in young people? A systematic review and meta-analysis". In: *Psychology and Psychotherapy: Theory, Research and Practice* 90.4, pp. 770–796.

Ubochi, Ebubechukwu Onyekachi Aham (2019). "Apple's Screen Time: The Utility of Digital Wellness Software in Curbing Smartphone Dependency". In.

Weisel, Kiona K. et al. (2019). "Standalone smartphone apps for mental health—a systematic review and meta-analysis". English. In: *NPJ Digital Medicine* 2.1. DOI: 10.1038/s41746-019-0188-8.

Wenzel, Amy (2017). "Basic strategies of cognitive behavioral therapy". In: *Psychiatric Clinics* 40.4. pmid:29080588, pp. 597–609.

Witkiewitz, Katie et al. (2014). "Development and evaluation of a mobile intervention for heavy drinking and smoking among college students." In: *Psychology of Addictive Behaviors* 28.3, p. 639.

Wubbolding, R. (2016). "Reality therapy". In: *Contemporary theory and practice in counseling and psychotherapy*, pp. 173–200.

Yao, Yuan-Wei et al. (2017). "Combined reality therapy and mindfulness meditation decrease intertemporal decisional impulsivity in young adults with Internet gaming disorder". In: *Computers in Human Behavior* 68. ID: 271802, pp. 210–216.

DOI: 10.1016/j.chb.2016.11.038. URL: https://www.sciencedirect.com/science/article/pii/S0747563216307798.

Zajac, Kristyn et al. (2017). "Treatments for Internet gaming disorder and Internet addiction: A systematic review". English. In: *Psychology of Addictive Behaviors* 31.8. Source type: Scholarly Journals; Object type: Feature; Object type: Article; Copyright: © 2017, American Psychological Association; AccNum: 2017-40993-001; CSAUnique: PSAR-2017-40993-001; DOI: 10.1037/adb0000315; ISSN: 0893-164X; ElecISSN: 1939-1501; PMID: 28921996; Peer Reviewed: true. DOI: 10.1037/adb0000315. URL: https://www.proquest.com/scholarly-journals/treatments-internet-gaming-disorder-addiction/docview/1940533247/se-2?accountid=15977%20https://discover.gcu.ac.uk/discovery/openurl?institution=44GLCU_INST%5C&vid=44GLCU_INST:44GLCU_VU2%5C&?url_ver=Z39.88-2004%5C&rft_val_fmt=info:ofi/fmt:kev:mtx:journal%5C&genre=article%5C&sid=ProQ:ProQ%5C%253Apsycarticles%5C&atitle=Treatments+for+Internet+gaming+disorder+and+Internet+addiction%5C%253A+A+systematic+review%5C&title=Psychology+of+Addictive+Behaviors%5C&issn=0893164X%5C&date=2017-12-01%5C&volume=31%5C&issue=8%5C&spage=979%5C&au=Zajac%5C%252C+Kristyn%5C%253BGinley%5C%252C+Meredith+K.%5C%253BChang%5C%252C+Rocio%5C%253BPetry%5C%252C+Nancy+M.%5C&isbn=%5C&jtitle=Psychology+of+Addictive+Behaviors%5C&btitle=%5C&rft_id=info:eric/2017-40993-001%5C&rft_id=info:doi/10.1037%5C%252Fadb0000315%20https://doi.org/10.1037/adb0000315.

Appendix